Note

Two integers a and b are congruent modulo m, written as $a \equiv b \pmod{m}$, if they leave the same remainder when divided by m.

Thinking Corner

How many integers exist which leave a remainder of 2 when divided by 3?

2.5.3 Modulo operations

Similar to basic arithmetic operations like addition, subtraction and multiplication performed on numbers we can think of performing same operations in modulo arithmetic. The following theorem provides the information of doing this.

Theorem 5

a, b, c and d are integers and m is a positive integer such that if $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then

(i) $(a + c) \equiv (b + d) \pmod{m}$ (ii) $(a - c) \equiv (b - d) \pmod{m}$

(iii) $(a \times c) \equiv (b \times d) \pmod{m}$

Illustration 3

If $17 \equiv 4 \pmod{13}$ and $42 \equiv 3 \pmod{13}$ then from theorem 5,

(i) $17 + 42 \equiv 4 + 3 \pmod{13}$

$59 \equiv 7 \pmod{13}$

(ii) $17 - 42 \equiv 4 - 3 \pmod{13}$

$-25 \equiv 1 \pmod{13}$

(iii) $17 \times 42 \equiv 4 \times 3 \pmod{13}$

$714 \equiv 12 \pmod{13}$

Theorem 6

If $a \equiv b \pmod{m}$ then

(i) $ac \equiv bc \pmod{m}$ (ii) $a \pm c \equiv b \pm c \pmod{m}$ for any integer c

Progress Check

1. The positive values of k such that $(k-3) \equiv 5 \pmod{11}$ are _____.
2. If $59 \equiv 3 \pmod{7}$, $46 \equiv 4 \pmod{7}$ then $105 \equiv$ _____ $\pmod{7}$, $13 \equiv$ _____ $\pmod{7}$, $413 \equiv$ _____ $\pmod{7}$, $368 \equiv$ _____ $\pmod{7}$.
3. The remainder when $7 \times 13 \times 19 \times 23 \times 29 \times 31$ is divided by 6 is _____.

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